

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-0954

Donegal 138 kV
March 21, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0954. The Interconnection Customer proposes a 103.3 MW solar photovoltaic facility interconnecting at the Donegal 138 kV substation in the Pennsylvania area. The request was received on March 01, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 261 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$701,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: MELROSE to UNIVERSAL STUDIOS TAP 2 138.0 kV
Contingency: ENO SUBSTATION to MELROSE 138.0 kV
Loading: 103.2% (Rating: 241 MVA)
DFAX: 33.0%
Pre-project loading: 89.1%
2. Facility: UNIVERSAL STUDIOS TAP 2 to LA CANADA 138.0 kV L
Contingency: ENO to MELROSE 138.0 kV
Loading: 100.1% (Rating: 229 MVA)
DFAX: 33.0%
Pre-project loading: 85.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MELROSE to UNIVERSAL STUDIOS TAP 2 SUBSTATION 138.0 kV	Reconductor line	\$662,000	14% (\$96,000)
UNIVERSAL STUDIOS TAP 2 to LA CANADA 138.0 kV L	Reconductor line	\$4,176,000	14% (\$605,000)
	Total allocated to project		\$701,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.